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In [ ]: #Practical No.1
        #Shreya Habib      Roll No.06
```

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: data = {'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Emily'],
'Age': [25, 30, 22, 35, 28],
'Salary': [50000, 60000, 45000, 70000, 55000]}
```

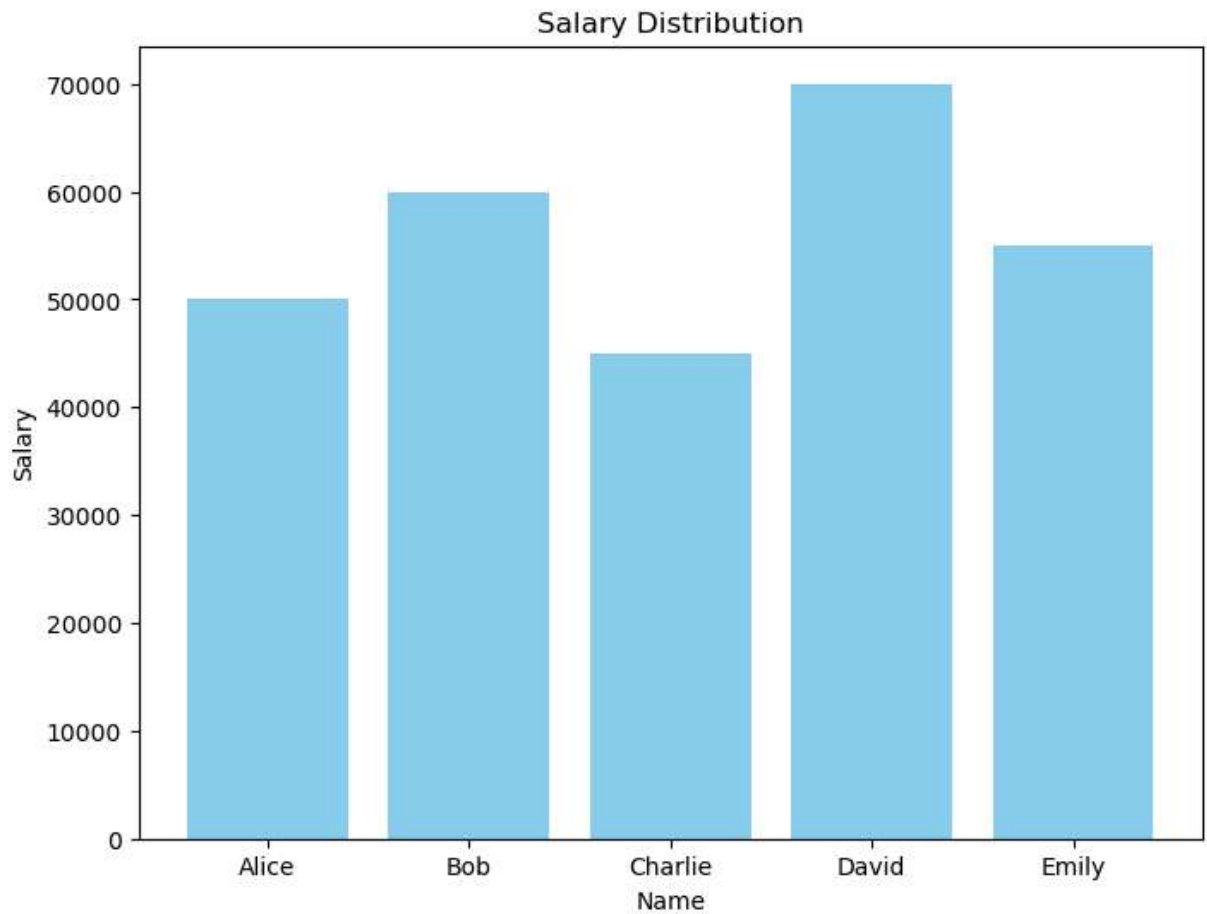
```
In [3]: df = pd.DataFrame(data)
```

```
In [4]: print("DataFrame:")
print(df)
```

DataFrame:

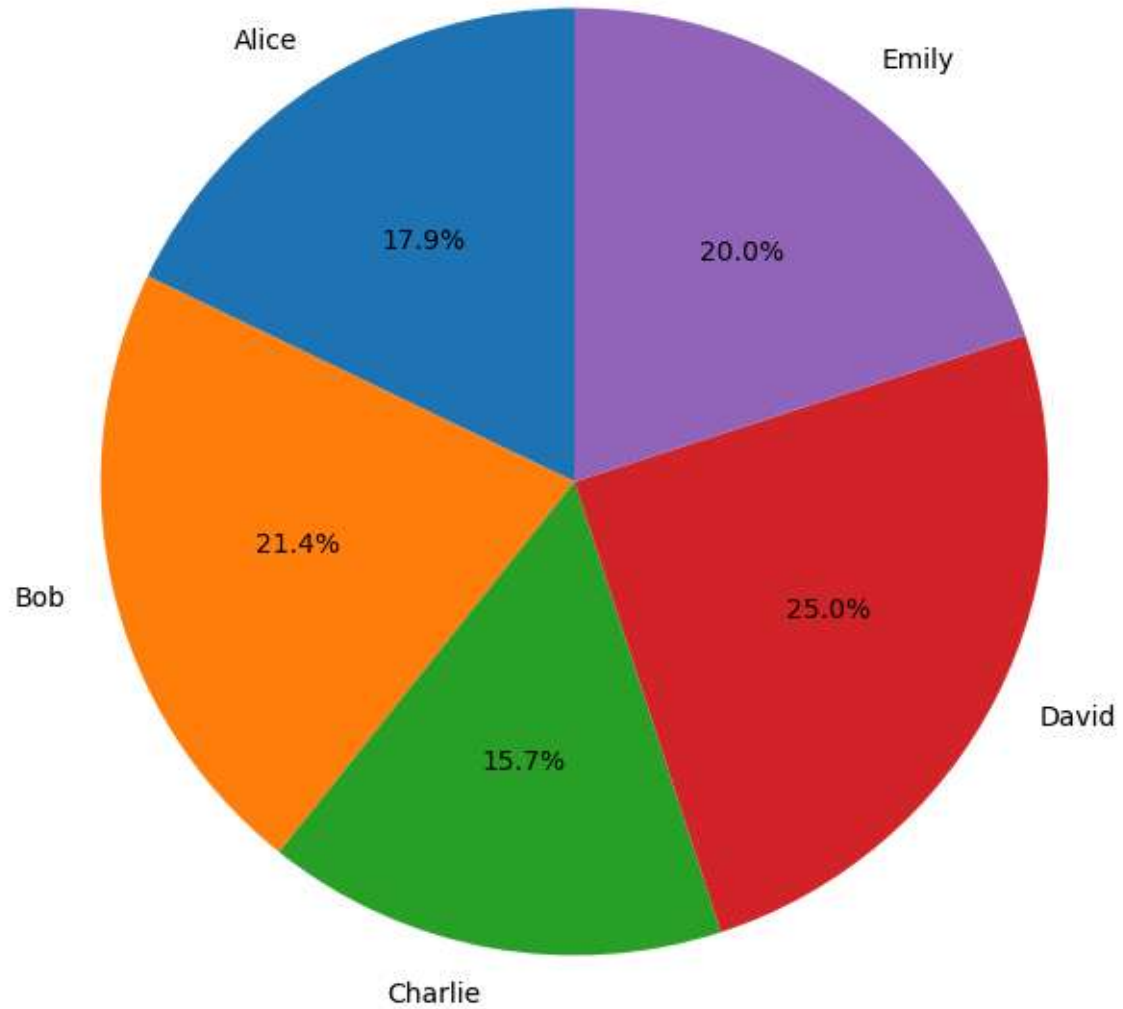
	Name	Age	Salary
0	Alice	25	50000
1	Bob	30	60000
2	Charlie	22	45000
3	David	35	70000
4	Emily	28	55000

```
In [5]: #plotting a bar chart for salary
plt.figure(figsize=(8, 6))
plt.bar(df['Name'], df['Salary'], color='skyblue')
plt.title('Salary Distribution')
plt.xlabel('Name')
plt.ylabel('Salary')
plt.show()
```



```
In [6]: #Plotting a pie chart for age
plt.figure(figsize=(8, 8))
plt.pie(df['Age'], labels=df['Name'], autopct='%1.1f%%', startangle=90)
plt.title('Age Distribution')
plt.show()
```

Age Distribution



```
In [9]: #Numpy
np.array([1,2,3])    #One dimensional
np.array([(1,2,3),(4,5,6)])  #Two dimensional
```

```
Out[9]: array([[1, 2, 3],
               [4, 5, 6]])
```

```
In [10]: np.zeros(3)
```

```
Out[10]: array([0., 0., 0.])
```

```
In [11]: np.ones((3,4))
```

```
Out[11]: array([[1., 1., 1., 1.],
                [1., 1., 1., 1.],
                [1., 1., 1., 1.]])
```

```
In [13]: np.eye(5)
```

```
Out[13]: array([[1., 0., 0., 0., 0.],
                [0., 1., 0., 0., 0.],
                [0., 0., 1., 0., 0.],
                [0., 0., 0., 1., 0.],
                [0., 0., 0., 0., 1.]])
```

```
In [15]: np.linspace(0,100,6)
```

```
Out[15]: array([ 0., 20., 40., 60., 80., 100.] )
```

```
In [17]: #pandas
df.head(5)
```

```
Out[17]:
```

	Name	Age	Salary
0	Alice	25	50000
1	Bob	30	60000
2	Charlie	22	45000
3	David	35	70000
4	Emily	28	55000

```
In [18]: df.tail(2)
```

```
Out[18]:
```

	Name	Age	Salary
3	David	35	70000
4	Emily	28	55000

```
In [20]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Name    5 non-null         object
1   Age     5 non-null         int64
2   Salary  5 non-null         int64
dtypes: int64(2), object(1)
memory usage: 252.0+ bytes
```

```
In [21]: df.describe()
```

Out[21]:

	Age	Salary
count	5.000000	5.000000
mean	28.000000	56000.000000
std	4.949747	9617.692031
min	22.000000	45000.000000
25%	25.000000	50000.000000
50%	28.000000	55000.000000
75%	30.000000	60000.000000
max	35.000000	70000.000000

```
In [ ]: !sudo apt-get install texlive-xetex texlive-fonts-recommended texlive-generic-recom
```

```
In [ ]:
```